What the Vocabulary Reveals: An Empirical Characterisation of the MRAD Corpus Behind the MedNarrate Medical Report Explainer

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TABLE I
MEMBER DETAILS

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***Abstract*—MRAD (Medical Report Analysis Dataset) is the 123,371-record corpus we assembled from three public sources to develop MedNarrate, an application that explains lab and pathology reports to patients. A report-type classifier trained on it scores a macro F1 of 1.00 on 12,454 reports from unseen patients. This paper asks what that score is made of, and what the corpus can and cannot test in the pipeline built on it. The raw CSV is not part of the released code, so we characterise the corpus through what is released: the split metadata, the evaluation metrics and, most usefully, the fitted TF-IDF vectoriser, whose inverse document frequencies can be inverted exactly into document counts for all 10,000 retained features. The training split is 81.2% blood-test reports, 11.0% health-record reports and 7.8% TCGA pathology reports. The blood and health-record reports are rendered from tables with fixed panels, 744 features are underscore-joined column names, and 406 features are 12-to-19-digit fractions left by unrounded unit conversion. The classifier separates the sources by format alone; its strongest TCGA features are "the", "of" and "and". Because MRAD rarely carries reference ranges, we then ran an ablation study of the MedNarrate pipeline on separate test sets. Asked to flag values against a printed range, a 1.5-billion-parameter language model was right 16 times in 30, against no wrong flags from the rule-based flagger. The ablation also exposed two faults in the output verifier, and showed that the verifier, which only asks whether a number exists in the source, would pass an unrounded MRAD value straight to a patient. We conclude that MRAD is a development corpus, not a benchmark, and set out remediation steps.**

***Index Terms*—Medical report explanation, dataset characterisation, corpus analysis, TF-IDF, shortcut learning, data quality, ablation study, small language models, clinical NLP, retrieval-augmented generation.**

I. INTRODUCTION

Patients increasingly receive lab and pathology reports before they see the doctor who ordered them, and those reports are written for clinicians. MedNarrate is our attempt to close that gap. It extracts the values from an uploaded report, decides with fixed rules whether each value is high or low, and only then asks a language model to explain the result in plain language, in English, Hindi or Telugu. Work on simplifying radiology reports with large language models has shown that the simplified text is usually readable and usually right, but not always [1], [2]. Keeping the numbers away from the model is our way of dealing with the "not always".

A pipeline like this needs data to be built against. We could not find a single public dataset that covered blood tests, routine health checks and pathology together, so we combined three public sources into one CSV with a common schema and called it MRAD. We then trained a simple report-type classifier on it, mostly as a sanity check, and it scored perfectly. A perfect score on a three-way text classification task is rarely good news. It usually means the task is easier than it looks, and in medical imaging there is a well-known precedent of models that learned which hospital an image came from instead of what it showed [3], [4].

This paper characterises MRAD in the way the Datasheets for Datasets proposal recommends [5], with one practical twist. The raw CSV is not in our code repository, so we work from the artefacts that are: a metadata file, a metrics file and the fitted vectoriser and model. Section IV shows that the vectoriser alone is enough to recover exact document counts for every retained feature, which turns out to be a surprisingly detailed view of the corpus.

We ask five questions. (i) What is MRAD made of, and in what proportions? (ii) What do the reports look like, at the level of their recurring tokens? (iii) What did the classifier learn, and does its perfect score mean anything? (iv) What do the answers imply for the parts of MedNarrate that will see real reports? (v) For the parts MRAD cannot test, how much does each safeguard in the pipeline contribute? Our contribution is a quantitative profile of the corpus, two data-quality defects that the documentation does not mention, an ablation study of the pipeline with a small-model comparison, and a short list of fixes ordered by how much risk they remove.

II. Related Work

A. Plain-language explanation of medical text

Jeblick et al. had radiologists rate ChatGPT simplifications of radiology reports and found them mostly correct and complete, with some wrong statements, missed findings and potentially harmful passages [1]. Lyu et al. found that prompt design reduces but does not remove information loss [2]. Rewriting 50 discharge summaries with GPT-4 lowered the reading grade from 11.0 to 6.2, yet 18 of 100 physician reviews raised safety concerns, mostly omissions [6]. Adapted models can match medical experts on clinical summarisation [7]. The common thread is that the errors left over are hard for a lay reader to detect, which motivates checking the facts outside the model.

B. Dataset documentation and shortcut learning

Gebru et al. argue that every dataset should ship with a datasheet describing its composition and collection [5]. Shortcut learning, where a model succeeds on a benchmark by exploiting a feature that does not carry over, is documented across vision and language [3]; Zech et al. showed a pneumonia detector picking up the hospital of origin [4]. The TCGA-Reports resource that supplies MRAD's pathology class was itself released as a machine-readable benchmark [8]. Our study applies the datasheet questions after the fact, from trained artefacts, and finds a source-level shortcut of the kind Geirhos et al. describe.

C. Grounded generation and small models

Retrieval-augmented generation puts source passages in the prompt [9] and improves medical question answering [10], but it does not stop a model from misreading what it retrieves, and hallucination remains a general property of neural generation [11]. Small language models of up to a few billion parameters can run on a phone or laptop and keep data local [12], [13]. Work on healthcare information systems has focused on the provider side, for example trust networks of doctors, departments and hospitals that treat the patient as an external entity [14]. We found no study that measures, component by component, how much each safeguard in a patient-facing report explainer contributes, or whether a small offline model can take the hosted model's place.

III. Dataset Overview And Design Intent

A. Purpose

MRAD was built to support development of the MedNarrate pipeline: training and testing a report-type classifier, exercising the lab value extractor and normaliser on many report shapes, and providing examples for the explanation prompts. It was not built as a benchmark for clinical accuracy, and nothing in it carries a clinician's judgement of what a patient should be told.

B. Sources

The corpus combines a general health-records dataset, a blood-test dataset and pathology reports from The Cancer Genome Atlas, the latter in the machine-readable form released as TCGA-Reports [8]. Each source had its own columns and naming, so every record was mapped to a common schema of 15 attributes. Patient identifiers were removed or replaced with generated keys, and medical terms were made consistent across the sources.

C. Logical schema

Table II lists the attributes as described in our dataset documentation. Four of them are referenced by name in the training code (Report_ID, Patient_ID, Report_Text and Source); for the rest we give the role only. The structured lab-value attribute is the one MedNarrate cares most about, but the classifier uses only the report text, with header lines such as "Report Type: Blood Test" stripped first so the label cannot be read off directly.

TABLE II
Logical Schema Of MRAD (15 Attributes)

Attribute	Role
Report_ID	Unique report key
Patient_ID	Generated patient key; basis of the split
Report type	Document category
Source	Blood, Health or TCGA; the classifier target
Age, gender	Demographics (two attributes)
Report date	Date of the report
Visit details	Encounter information
Report_Text	Full report text; the model input
Quality score	Numeric text-quality score
Quality category	Banded version of the score
Diagnosis	Recorded diagnosis, where present
Label	Task label
Structured lab values	Values parsed from the text

The four names with underscores are those used in the training code; the others are described by role.

D. Cleaning and splits

The loader drops rows with empty text and removes exact duplicates on the pair (patient, text). Of 123,371 raw records, 4 were duplicates, leaving 123,367 usable reports from 110,677 patients. The split is by patient, not by report: patients are shuffled with seed 42 and divided 80/10/10, and the code checks that no patient appears in two splits. This gives 98,492 training, 12,421 validation and 12,454 test reports. The documentation reports these totals but not the class make-up of any split, which is the first thing we recover below.

IV. Proposed Method

A. The MedNarrate pipeline

MedNarrate reads a PDF or photo with PyMuPDF or OCR, rejects documents without enough medical evidence, tags entities with a biomedical NER model, and extracts each test, value, unit and printed range with three rule parsers tried in a fixed order. Each value is flagged by comparing it with its printed range; with no range it is left "not classified". The report is de-identified, chunked and indexed for retrieval, and a hosted language model (Gemini 3.1 Flash-Lite) writes a clinician summary and a patient summary. A verifier checks every number and a list of serious diagnoses against the source, and translations are rejected if any number changes. The companion case study describes each stage [15]. Here, the

method has two parts: a characterisation of MRAD from its released artefacts (Sections IV-B to IV-E), and an ablation of the pipeline (Section IV-F).

B. Evidence

Our corpus evidence comes from three places. The first is the dataset documentation and the loading code, which state the design intent and the cleaning rules. The second is the released artefacts: metadata.json and metrics.json, with the split sizes, test confusion matrix and per-class scores, and the fitted scikit-learn [16] TF-IDF vectoriser and logistic-regression model. The third is a set of small experiments we ran on the released code at commit ca69541, described in Section V-D.

C. Recovering document frequencies

The vectoriser keeps the 10,000 most frequent word unigrams and bigrams (minimum document frequency 2, default token pattern of two or more word characters, lower-cased). With smooth_idf enabled, scikit-learn stores for each feature

$$idf(t) = ln[(1 + n) / (1 + df(t))] + 1, \quad (1)$$

where n is the number of training documents and df(t) the number containing t. Since n = 98,492 is known from the metadata, the relation inverts to

$$df(t) = (1 + n) \cdot exp(1 - idf(t)) - 1. \quad (2)$$

Applied to all 10,000 stored weights, (2) returns values within 1.5×10^{-11} of an integer, so the recovery is exact up to floating-point error. This gives us, for every retained feature, the number of training reports that contain it. It does not give us features that fell outside the 10,000, and it says nothing about the validation or test splits.

D. Inferring class sizes and probing the classifier

If a set of features appears in every report of one source and in no report of the others, their document frequency equals that source's size. We look for plateaus, many features sharing exactly the same df, check whether the plateau values add up to n, and cross-check against the test-split class counts from the confusion matrix. To probe the classifier we read off the ten largest positive weights per class, the share of each class's positive weight that falls on purely numeric features and on underscore-joined tokens, and the model's predictions on nine short documents written in formats absent from MRAD.

E. Limits of the instrumentation

Everything below the frequency cut-off of the vocabulary is invisible to us, so rare phenomena cannot be measured. And the training split is 80% of patients chosen at random, so proportions measured on it should transfer to the full corpus, but exact counts do not.

F. Ablation and model-comparison protocol

MRAD cannot test the flagging step (Section V-B), so the pipeline ablation uses separate test sets: 42 gold values in five report layouts, 30 printed result lines with known flags, 15 questions over a synthetic report, 56 translations into seven Indian languages, 203 deliberately edited summaries, 26 identifiers in 12 synthetic report headers and 60 chat messages. Following the usual protocol [17], each run changes one component and keeps the rest fixed. For the model comparison, Qwen2.5-1.5B-Instruct [13] in Q4_K_M quantisation runs through llama.cpp [18] on four CPU threads with the app's own prompts, temperature 0.2 and seed 42, and is scored with the same code as the hosted chain. Translation quality is measured with chrF [19]. Scripts, data and raw model replies are in the repository under mednarrate-backend/experiments/review2.

V. Results And Discussion

A. Corpus composition

Fig. 1 shows the recovered document frequencies in rank order. The curve is not the smooth decline usually seen in text. It is a staircase with four flat steps. Twenty-nine features, among them "µl", "hemoglobin", "mchc" and "direct_bilirubin", occur in exactly 90,840 reports. Thirteen, including "aptt", "free_t4" and "pmol", occur in exactly 79,984. A block of 111 occurs in exactly 10,856. Since 79,984 + 10,856 = 90,840, and 98,492 − 90,840 = 7,652, the steps are the class sizes: 79,984 blood reports, 10,856 health-record reports and 7,652 pathology reports. A handful of features sit one to six reports above the blood plateau ("inr" and "fibrinogen" at 79,985, "ldh" at 79,990); the most likely explanation is that a few pathology reports mention them.

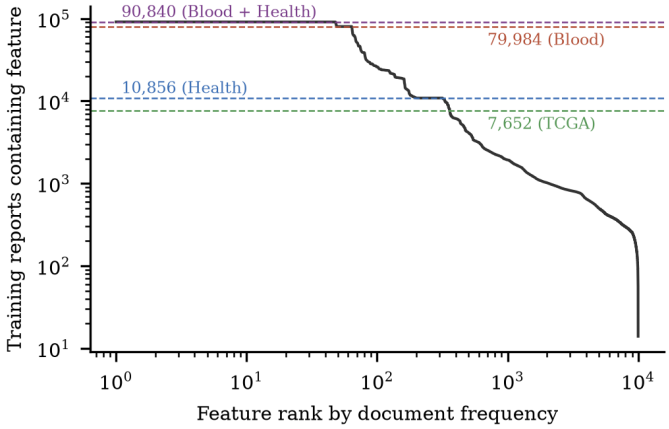


Fig. 1. Training reports containing each of the 10,000 vocabulary features, in rank order. The flat steps coincide exactly with the source sizes; the step at 90,840 is the analyte vocabulary shared by blood and health-record reports.

Table III and Fig. 2 put these numbers next to the test split. The proportions agree to within 1.1 percentage points, which is what a random patient-level split should produce, and which supports reading the plateaus as class sizes.

TABLE III
MRAD Composition By Source

Source	Training	Share	Test	Share
Blood	79,984	81.2%	10,028	80.5%
Health	10,856	11.0%	1,512	12.1%
TCGA	7,652	7.8%	914	7.3%
Total	98,492	100%	12,454	100%

Training counts are recovered from the vectoriser (Section IV-D); test counts come from the released confusion matrix.

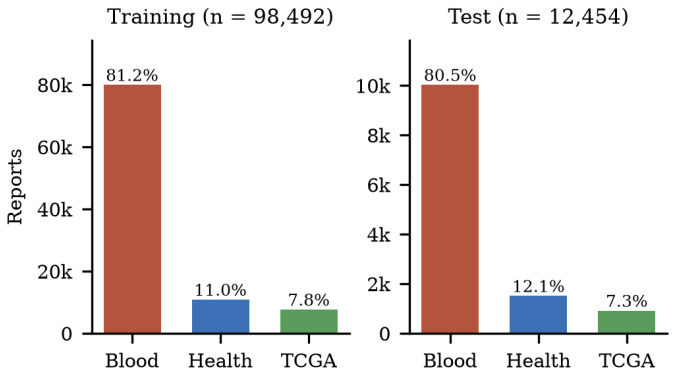


Fig. 2. Source composition of the training and test splits. Blood-test reports make up about four in five reports in both.

The imbalance matters for anything trained on MRAD. A model that always answered "Blood" would already be 80% accurate, and the pathology class, the only one written in prose, has fewer than one report in twelve. The health-record source shows a slightly higher share in the test split than in training. With a patient-level split this happens when one source has more reports per patient, and the gap between 123,367 reports and 110,677 patients (about 12,700 repeat reports) has to sit somewhere. We cannot say where from the artefacts alone.

B. What the reports look like

Fixed panels rendered from tables. A step in Fig. 1 means that a whole set of terms appears in every single report of a source. For the blood source this set is a full panel: the complete blood count with red-cell and platelet indices (MCV, MCH, MCHC, RDW, MPV, PDW), the five-part differential, liver enzymes (ALT, AST, ALP, GGT), bilirubin fractions, calcium and phosphorus, TSH and free T4, LDH, and a coagulation block of INR, aPTT and fibrinogen. Real laboratories do not order all of these for every patient. The health-record source has its own fixed panel of 63 single-word features at exactly 10,856, covering lipids and their ratios, iron studies, glucose and HbA1c, renal markers, electrolytes and 19 urine fields from colour and appearance to casts and crystals.

The names themselves give away how the text was produced. In total, 744 of the 10,000 features contain an underscore ("total_cholesterol", "neutrophils_percent", "urine_specific_gravity", "a_g_ratio"). Those are database column names, not the way any printed lab report labels a test. The blood and health-record reports in MRAD are, in other words, table rows serialised as "name : value unit" lines. The one MRAD-style sample kept in our test suite ("Calcium : 9.675 mg/dL / AST : 21.69 U/L / ALT : 28.63 U/L") fits that description and carries no reference range. The words "reference", "ref", "interval" and "units" are not among the 10,000 features at all, which suggests that reference ranges are rare or absent in the serialised reports, though we cannot rule out ranges written as bare number pairs.

Unrounded floating-point values. The second defect is less visible. 605 features contain a run of twelve or more digits, and 406 of them are not also column-name tokens (Fig. 3). Because the token pattern splits on the decimal point, a value such as 0.08128654970760232 becomes the token "08128654970760232", and the bigram "08128654970760232 mg" shows it is followed by a mass unit. This particular token appears in 549 reports. Many of these fractions are exact multiples of 1/17.1 (0.0058479532… is 0.1/17.1), and 17.1 is the factor that converts bilirubin from µmol/L to mg/dL. Other runs such as "999999999999" next to "µl" are classic binary floating-point residue. Our reading is that one source stored values in SI units, they were converted during harmonisation, and the results were written into the report text without rounding.

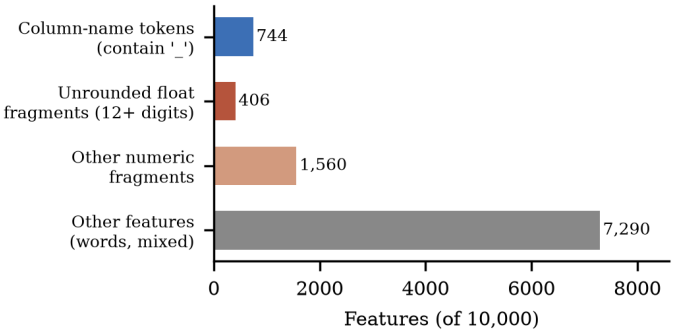


Fig. 3. Composition of the 10,000-feature vocabulary. Categories are exclusive and assigned top to bottom. Nearly one feature in ten is either a column name or a fragment of an unrounded float.

For a classifier this is noise. For MedNarrate it is a real problem: the patient prompt tells the model to quote exact values, and the grounding check accepts any number found in the source, so an unrounded value would be faithfully passed on to a patient as "0.08128654970760232 mg/dL".

Pathology prose. The TCGA reports are the opposite: ordinary clinical prose with no fixed panel. "of" appears in 97% of them, "diagnosis" in about 90%, "tumor" in 89%, "specimen" in 76% and "carcinoma" in 68%. Of the 1,362 purely alphabetic single-word features, 1,297 occur in 7,652 reports or fewer, which is consistent with almost all ordinary vocabulary in MRAD coming from this one source.

C. What the classifier learned

Table IV gives the released test results. All 12,454 test reports are classified correctly, and one of 12,421 validation reports is wrong (validation accuracy 0.99992).

TABLE IV
REPORT-TYPE CLASSIFIER ON THE TEST SPLIT

Class	Support	Precision	Recall
Blood	10,028	1.000	1.000
Health	1,512	1.000	1.000
TCGA	914	1.000	1.000
Macro avg.	12,454	1.000	1.000

TF-IDF (unigrams and bigrams, 10,000 features) with logistic regression, C = 1.0; header lines removed before vectorising.

Given Section V-B, this is unsurprising. Fig. 4 shows the largest positive weights. For Blood, the single biggest weight is on "µl", followed by tokens that only the blood panel contains ("pmol", "free_t4", "seconds fibrinogen"). For Health, the weights sit on units and qualitative urine values ("µg dl", "mmol", "negative", "nil", "trace"). For TCGA, the top three features are "the", "of" and "and". Summing positive weights

by feature type makes the point more sharply: 26% of the Blood class's positive weight lies on bare number fragments such as "13" and "31" (pieces of typical haemoglobin and MCH values), and 42% of the Health class's positive weight lies on underscore column names. Neither is medical knowledge.

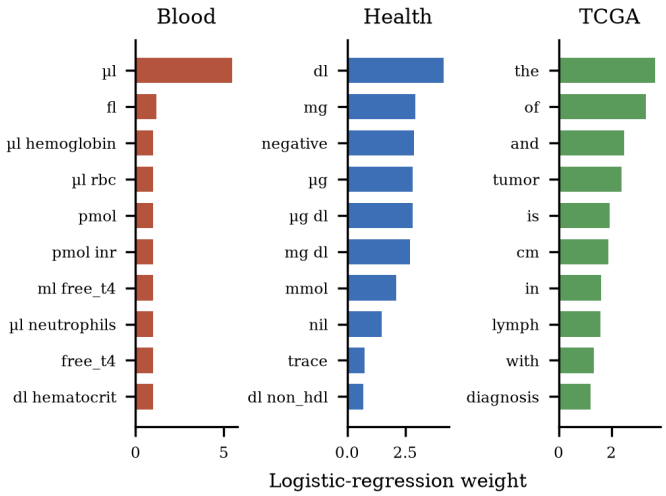


Fig. 4. The ten largest positive logistic-regression weights per class. The model separates the sources by units, panel membership and the presence of English function words.

Header stripping was meant to stop the label leaking into the features, and it does remove the literal label. But the label is also encoded in the format of the whole document, and no amount of header removal changes that. This is the shortcut-learning pattern described by Geirhos et al. [3]: the model has found the easiest feature that predicts the label in this data, and that feature does not carry over.

D. Behaviour outside MRAD

To see what the classifier does with reports that do not look like MRAD, we wrote nine short documents (Fig. 5). The MRAD-style control is classified as Blood with 0.96 confidence. A printed Indian complete blood count is also called Blood (0.77), which is right, and so is a thyroid panel (0.68). After that the errors are systematic. A lipid profile, the content of the Health class, is called Blood with 0.81. A urine routine report, also Health content, goes to TCGA (0.52). A chest X-ray report (0.71) and a discharge summary (0.93) go to TCGA because they are written in sentences, and a one-line non-medical sentence about getting a sugar test gets Blood with 0.56. With three closed classes and no rejection option, every input gets a confident-looking label.

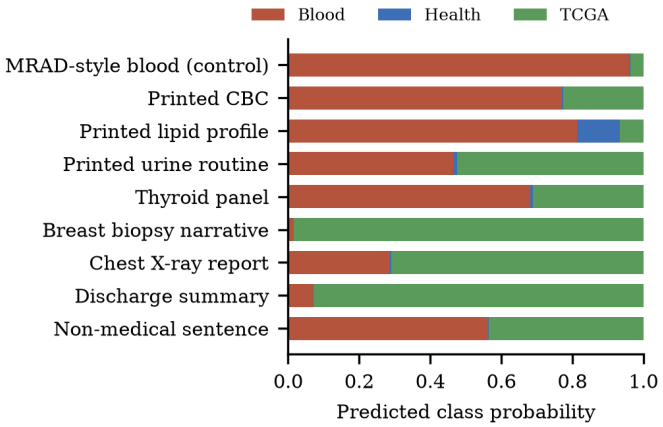


Fig. 5. Class probabilities for nine documents in formats absent from MRAD. Prose of any kind is pulled toward TCGA; lab content without MRAD's column names is pulled toward Blood.

MedNarrate's extractor is rule-based first and only falls back to a language model when the rules find nothing. We wrote nine reports in six layouts with 35 values to see how MRAD's format compares with printed reports. With the model fallback disabled, the rules found all 4 values in the MRAD-style layout (L4) and all 24 in the three printed one-line and vertical layouts (L1 to L3), with one spurious value in L3. A table with an extra method column (L5) produced the right numbers under the wrong names, and a narrative sentence (L6) produced nothing, as intended (Fig. 6).



Fig. 6. Rule-based lab extraction by layout on 35 hand-written values. L4 is the MRAD format; L1 to L3 are common printed formats; L5 and L6 fall outside the rules.

Two points follow for MRAD. All four L4 values were found, but every one was labelled "not classified", because the format has no reference range and the flagging logic refuses to guess one. If most MRAD blood and health-record reports lack ranges, as Section V-B suggests, then MRAD cannot exercise the part of MedNarrate that decides high and low, which is arguably its most important part. And the extraction layouts that matter for real users (L1 to L3, scans and phone photos) are not represented in MRAD at all; we had to write them ourselves.

E. Discussion

The findings separate cleanly. As an engineering resource MRAD did its job. It is large, it has stable keys, the patient-level split is implemented and checked, and it gave the pipeline a steady supply of inputs in three very different

shapes. As evidence, it is weak, and the reasons are structural rather than a matter of size. Two of the three sources are tables rendered as text with a fixed set of tests in every report, and they keep the fingerprints of that process: database column names and unrounded unit conversions. The third is real clinical prose, but from one programme and one kind of report. A model trained on this learns to tell the sources apart, which is all the perfect F1 measures. None of our three headline numbers, the 1.00 macro F1, the 0.99992 validation accuracy and the clean patient split, says anything about how the classifier or the pipeline would behave on an Indian lab's printed report or a phone photo of one.

We find it useful that all of this came out of artefacts most projects discard. A fitted vectoriser is usually treated as a by-product of training, but because its IDF weights encode exact document counts, it is in effect a compressed index of the corpus. Datasheets [5] ask authors to report composition and collection process; here the vectoriser documented both without being asked.

VI. Ablation Study

Because MRAD cannot test flagging, we measured the pipeline's safeguards directly. Table V summarises the ablations most relevant to the corpus findings; the companion case study reports all fifteen [15].

TABLE V
SELECTED ABLATIONS OF THE MEDNARRATE PIPELINE

Change	Test set	Full	Ablated
Column parser removed	42 values	R 81.0%	R 52.4%, 4 wrong
Line parser removed	42 values	R 81.0%	R 61.9%
Printed ranges deleted	12 values	12 correct flags	12 not classified
Flags decided by SLM	30 lines	26 right, 0 wrong	16 right, 14 wrong
Structured values removed (SLM)	39 values	82.1% cited	43.6% cited
Report context removed (SLM)	15 questions	10/10, 4/5 declined	3/10 flagged missing, 2/5 declined
Grounding verifier removed	203 edits	145 caught	0 caught
Translation number check removed	24 corrupted	24 rejected	0 rejected
De-identification removed	26 identifiers	4 leak	26 leak

R = recall. SLM = Qwen2.5-1.5B-Instruct, Q4_K_M, CPU. Each row changes one component; the rest is the full system.

Flags belong outside the model. With printed ranges deleted, all 12 extracted values were left "not classified", which is how the pipeline treats every MRAD value. When the small model was asked to judge 30 printed values instead, it got the 16 abnormal ones right and labelled all 14 normal ones low or high. The rules read 26 of the same lines and flagged every one correctly. This is the step MRAD cannot test, and it is the one where the gap between rules and a model is widest.

The verifier has two faults. On 203 deliberately edited summaries the grounding verifier caught 145. It accepts any number that appears inside a longer number in the source, so a glucose of 112 changed to 1120 passes when the report also prints 11200, and it ignores 0 to 5. Its diagnosis check is skipped whenever a summary contains "not provided", and the patient prompt requires a section headed "Information Not Provided", so the check never runs on real summaries: all six small-model summaries passed while five named a condition absent from the report. A related limit bears on Section V-B. The verifier asks only whether a number exists in the source, so an unrounded MRAD value is grounded by definition and would reach the patient unchanged. Rounding at serialisation (R1 below) is the only fix for that case.

A small model as an offline fallback. Table VI compares the hosted chain with Qwen2.5-1.5B on the same test sets. Given the report, the small model answered all ten report questions correctly and invented no numbers, with a median latency of 0.73 s on a laptop CPU. Its translations were poor: chrF on Hindi of 26.9 against 67.1, and one or more numbers lost or changed in 19% of the translations that contained one. It is usable for English questions over a report already in its context, not for translation or for deciding flags.

TABLE VI
HOSTED LLM CHAIN VERSUS SMALL LANGUAGE MODEL

Measure	LLM chain	SLM
Report questions correct	10/10	10/10
Unanswerable questions declined	5/5	4/5
Numbers kept in translation	100%	82.1%
chrF, Hindi	67.1	26.9
Degenerate translations	n/m	6/56
Median latency, questions	1.92 s	0.73 s
Model on device	none	1.12 GB
Works offline	No	Yes

LLM chain: Gemini 3.1 Flash-Lite primary with hosted fallbacks. n/m = not measured.

VII. Threats To Validity

First, the class sizes are inferred from feature plateaus, not counted. The inference rests on the assumption that the plateau features occur in every report of one source and in none of the others. The exact additive match (79,984 + 10,856 = 90,840) and the agreement with test proportions make an alternative explanation unlikely, but a count on the CSV would settle it.

Second, we only see the 10,000 most frequent features. Statements of the form "X does not occur" mean "X is not frequent enough to be retained", and that is how we have phrased them. In particular, the claim that reference ranges are rare rests on the absence of a few words and on one sample, and should be checked directly.

Third, the interpretation of the long fractions as bilirubin conversions rests on the factor 17.1 and the following "mg" token. Other conversions may also be involved, and we have not checked which analytes carry them.

Fourth, the probe documents, extraction reports and ablation test sets were written by us, are small, and were designed with knowledge of the code. They illustrate behaviour; they are not accuracy estimates. The hosted-model figures come from a run on 17 October 2026 and the small-model figures from 19 October 2026 on the same laptop, so latencies are not directly comparable: the hosted figures include the network round trip.

VIII. Conclusion And Future Work

MRAD is a 123,367-report corpus that is 81% blood-test reports, 11% health-record reports and 8% pathology reports.

Its two structured sources are table rows rendered as text, with a fixed panel of tests, database column names and, in places, unrounded unit conversions. A classifier trained on it scores perfectly because the sources are trivially separable by format, and it carries that shortcut to new documents, sending any prose to the pathology class. None of this is visible in the dataset's headline numbers, and all of it was recoverable from the fitted vectoriser. The ablation adds a second lesson: the part of MedNarrate that MRAD cannot test, deciding high and low, is the part where a language model is least reliable and fixed rules are most valuable.

The following steps are ordered by the risk they remove relative to the effort they need.

R1. Round values at serialisation. Round every converted value to the precision a laboratory would print and regenerate the report text, so a 17-digit number never reaches a patient.

R2. Replace column names with printed labels. Map "total_cholesterol" to "Total Cholesterol" and so on before rendering, which also removes the classifier's easiest shortcut.

R3. Add reference ranges and vary the panel. Attach a plausible laboratory range to each value and sample a realistic subset of tests per report. Without ranges, MRAD cannot test the flagging logic at all.

R4. Document the composition. State the 81/11/8 split and the per-source counts in the dataset documentation, and say plainly that two sources are synthesised from tables.

R5. Evaluate on out-of-format data. Keep a held-out set of printed, scanned and photographed reports, ideally de-identified real ones obtained with permission, and give the classifier a way to say "none of these".

R6. Fix the verifier. Compare whole numeric tokens only, test negation within the sentence that names a diagnosis, and keep the 203 edited summaries as regression tests.

R7. Keep the classifier diagnostic. Until R1 to R5 are done, use it as a probe of source separability. MedNarrate does not call it for any decision, and it should not start to.

Until then, the honest description of MRAD is a development corpus for the MedNarrate pipeline, not a benchmark for it. We plan to publish the cleaned corpus with a datasheet once R1 to R4 are complete.

References

[1] K. Jeblick et al., "ChatGPT makes medicine easy to swallow: an exploratory case study on simplified radiology reports," *Eur. Radiol.*, vol. 34, pp. 2817-2825, 2024.

[2] Q. Lyu et al., "Translating radiology reports into plain language using ChatGPT and GPT-4 with prompt learning: results, limitations, and potential," *Vis. Comput. Ind. Biomed. Art*, vol. 6, art. 9, 2023.

[3] R. Geirhos et al., "Shortcut learning in deep neural networks," *Nat. Mach. Intell.*, vol. 2, pp. 665-673, 2020.

[4] J. R. Zech et al., "Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study," *PLoS Med.*, vol. 15, no. 11, e1002683, 2018.

[5] T. Gebru et al., "Datasheets for datasets," *Commun. ACM*, vol. 64, no. 12, pp. 86-92, Dec. 2021.

[6] J. Zaretsky et al., "Generative artificial intelligence to transform inpatient discharge summaries to patient-friendly language and format," *JAMA Netw. Open*, vol. 7, no. 3, art. e240357, 2024.

[7] D. Van Veen et al., "Adapted large language models can outperform medical experts in clinical text summarization," *Nat. Med.*, vol. 30, pp. 1134-1142, 2024.

[8] J. Kefeli et al., "TCGA-Reports: A machine-readable pathology report resource for benchmarking text-based AI models," *Patterns*, vol. 5, no. 3, art. 100933, 2024.

[9] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, vol. 33, 2020, pp. 9459-9474.

[10] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking retrieval-augmented generation for medicine," in *Findings Assoc. Comput. Linguistics: ACL 2024*, 2024.

[11] Z. Ji et al., "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, art. 248, 2023.

[12] Z. Lu et al., "Small language models: Survey, measurements, and insights," arXiv:2409.15790, 2024.

[13] Qwen Team, A. Yang et al., "Qwen2.5 technical report," arXiv:2412.15115, 2024.

[14] A. Gayen, S. Chakraborty, J. Chakraborty, and A. Jana, "Trust@Health: A trust-based multilayered network for scalable healthcare service management," *IEEE Access*, vol. 13, pp. 167061-167079, 2025.

[15] G. M. Yoshitha, G. H. Chowdary, R. K. Teja, and D. J. Sandeep, "MedNarrate: Explaining medical lab and pathology reports to patients using RAG-based LLMs," 23CSE471 case study report, Amrita Vishwa Vidyapeetham, Coimbatore, 2026.

[16] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825-2830, 2011.

[17] R. Meyes, M. Lu, C. W. de Puiseau, and T. Meisen, "Ablation studies in artificial neural networks," arXiv:1901.08644, 2019.

[18] G. Gerganov et al., "llama.cpp: LLM inference in C/C++," GitHub repository. [Online]. Available: https://github.com/ggml-org/llama.cpp

[19] M. Popovic, "chrF: character n-gram F-score for automatic MT evaluation," in *Proc. 10th Workshop Statist. Mach. Transl.*, 2015, pp. 392-395.

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